



## Mikolaj Duchlinski

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### ABOUT ME

Artificial Intelligence graduate (BSc) from Vrije Universiteit Amsterdam (Intelligent Systems track), with a minor in Electronics for Robotics from TU Delft. Possessing a strong foundation in electrical engineering and high proficiency in advanced computational intelligence, I specialize in bridging the gap between physical hardware and intelligent algorithms. An expertise demonstrated in projects by designing evolutionary algorithms, training reinforcement learning models, and applying linear algebra and statistical methods to optimize autonomous systems.

I am highly proficient in object-oriented Python programming, algorithm design, and complexity analysis, with hands-on experience structuring and cleaning complex datasets via SQL and data wrangling techniques. My technical capabilities also extend to electronics, hardware assembly, and 3D printing. This hybrid software-hardware skill set was directly applied during a bachelor's thesis focused on optimizing drone robotics through evolutionary computation.

Great complement to my technical expertise is a strong international background—including an exchange at Universitat Politècnica de València and a Propedeuse in Business & Event Management. This foundation brings valuable project management, logical problem-solving, and cross-functional communication skills to technical engineering environments.

Currently, I am expanding my expertise in control system engineering, advanced automation and robotics through independent technical projects and advancing my proficiency in C++. I am actively seeking full-time engineering roles at the moment, with the intent to pursue a part-time Master's degree in Electrical Engineering or Automation & Robotics Engineering starting in September 2027. This part-time academic path is intentionally chosen so that I can fully dedicate myself to a full-time professional role, allowing me to deliver value to my employer while continuing to deepen my technical and academic foundation.

### SKILLS

#### Software Engineering & DevOps

Python (OOP, APIs) | C/C++ (basic) | Linux | Docker | Slurm | Prolog | Conda | SSH | tmux | Ubuntu server | Git/ GitHub | Computational Thinking | VS Code

#### AI & Machine Learning

Machine Learning | Evolutionary Algorithms | Computational Intelligence | Deep Learning | Multi-Agent Systems | Stable-Baseline3 | Gymnasium | PyTorch | Reinforcement Learning

#### Digital Systems & Hardware

VHDL | Quartus II | AMD Vivado | MATLAB | Questasim | Simulink | FPGA (Artix-7) | Raspberry Pi 5 | 3D Printing

#### Electrical Power Systems

Power Electronics | Battery Management | MPPT Algorithms | Finite State Machines | Circuit Analysis

#### Operations & Management

Operations | Stakeholder Communication | Resource Planning | Event/Operational Logistics | Project Management | Business Reporting

#### Data Management & Analytics

Data Wrangling | Knowledge Graphs | SQL | Database Design | Pandas | Numpy | Data Visualization (Seaborn, Matplotlib)

#### Mathematics & Algorithms

Linear Algebra and Calculus | Probability and Statistics | Data Structures and Algorithms | Automata and Complexity | Logic and Sets | Graphs and Search Algorithms

#### NLP & Text Mining

Natural Language Processing | Text Mining | Transformer architecture | OpenAI Whisper | Dialogflow

## PROJECTS

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01/07/2026 – CURRENT

### Autonomous Drone Navigation & Hardware Deployment

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**Context:** Independent Project | Vrije Universiteit Robotics Lab

- Constructing an **3 and 5-inch quadcopter** to establish a physical hardware deployment pipeline.
- **3-D printing** the drone's frame and other necessary components.
- Integrating an onboard **Raspberry Pi** single-board computer which runs pre-trained **PPO models** in real time for autonomous flight navigation in the environment of **OptiTrack**.
- Applying the autonomously navigating drone pipeline from a standard quadcopter frame onto custom, asymmetrical drone designs evolved during the Power-Aware Co-Optimization of Unconventional Hexacopters via Evolution and Learning research project.

**Key Technologies:** Raspberry Pi, Python, Real-Time RL Inference, Flight Controllers, Embedded Systems, 3D printing, OptiTrack.

01/04/2026 – 30/06/2026

### Power-Aware Co-Optimization of Unconventional Hexacopters via Evolution and Learning

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**Context:** Bachelor Thesis Project | Vrije Universiteit Amsterdam

**Grade:** 8.5 / 10 (15.0 ECTS)

- Formulated research questions and executed empirical experiments in evolutionary robotics and reinforcement learning simulator environment.
- Designed a bi-objective evolutionary framework using **NSGA-II** in Python to co-optimize physical hexacopter morphologies and neural network controllers trained via **Proximal Policy Optimization (PPO)** in PyTorch.
- Modeled realistic 4S LiPo battery dynamics (voltage sag, C-rating discharge limits) to evaluate energy efficiency alongside flight performance across figure-8 and shuttlerun navigation tracks.
- Executed parallelized evolutionary compute loops on a cluster utilizing 192 CPU cores.
- Achieved up to a **31.6% reduction** in energy consumption while maintaining **95.7% of baseline flight performance**.

**Key Technologies:** Python, PyTorch, Reinforcement Learning (PPO), Evolutionary Algorithms (NSGA-II), Battery Physics Modeling, High-Performance Computing (Cluster), Stable-Baseline3.

01/06/2026 – 30/06/2026

### Pulse - personalized content feed

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**Context:** Applied Programming for AI | Vrije Universiteit Amsterdam

**Grade:** 8.5 / 10 (6.0 ECTS)

- Architected a full-stack personalized content feed platform using **Python**, **Flask** (Application Factory pattern, Blueprints), and **SQLite**, decoupling heavy content processing from a lightweight frontend via **RESTful JSON APIs**.
- Engineered a pipeline with integration of **GNews**, **YouTube Data API v3**, and **Lemmy REST APIs**, normalizing heterogeneous payloads into unified data schemas with server-side caching to respect API limitations.
- Developed an **NLP** enrichment and dynamic ranking engine combining vaderSentiment scoring, keyword matching, temporal freshness decay, and user feedback loops (upvotes/downvotes).
- Implemented a cross-source clustering algorithm to group related news articles, videos, and forum threads of identical events into single multi-perspective stories.
- The project follows rules of code reliability by implementing comprehensive type hints, writing a test suite of 325 unit and integration tests using pytest with mocked external dependencies.

**Key Technologies:** Python, Flask, RESTful APIs (YouTube Data API v3, GNews, Lemmy), SQLite, vaderSentiment, Pytest, Token-based Authentication, JavaScript, HTML/CSS.

01/11/2025 – 31/01/2026

### Autonomous Solar-Powered Mars Rover

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**Context:** Hardware Systems Project | Technische Universiteit Delft

**Grade:** 8.5 / 10 (5.0 ECTS)

- Designed and synthesized a complete digital control system in VHDL on an AMD Xilinx Basys3 Artix-7 FPGA to navigate an autonomous mobile robot.
- Implemented **Finite State Machines (FSMs)**, **PWM motor drivers**, **synchronous input buffers**, and line-tracking & finding logic.
- Engineered a physical **power management system** featuring a buck converter, **supercapacitor energy buffer**, solar panel array, and a custom analog Schmitt trigger comparator with hysteresis.

**Key Technologies:** VHDL, AMD Vivado, FPGA (Artix-7), FSM Design, PWM Control, Buck Converters, Power Electronics.

01/01/2026 – 31/01/2026

### Voice-Driven Conversational AI Pipeline

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**Context:** Applied AI Project | Vrije Universiteit Amsterdam

Grade: 9.5 / 10 (6.0 ECTS)

- Engineered an end-to-end voice-interactive recipe recommendation system.
- Fine-tuned a **BERT** transformer model for simultaneous **intent classification** and **slot filling**.
- Integrated the custom classifier with a **MARBEL (GOAL language)** dialogue manager, a **Prolog knowledge base** for structured recipe retrieval, and OpenAI **Whisper** for Automatic Speech Recognition (ASR) and Text-to-Speech (TTS).
- Hosted the whole system locally on a self-created website, using **JavaScript**, **HTML** and **CSS**.

**Key Technologies:** Natural Language Processing (NLP), BERT, PyTorch, Dialogflow, Prolog, OpenAI Whisper, JavaScript, HTML, CSS.

01/02/2025 – 31/03/2025

**Deep Learning Architecture**

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**Context:** Machine Learning Project | Vrije Universiteit Amsterdam  
**Grade:** 8.5 / 10 (6.0 ECTS)

- Constructed a **LeNet-5 CNN** architecture from scratch using pure **NumPy**, achieving a 12x execution speedup via **Numba JIT compilation**.
- Implemented a **ResNet-18** model in PyTorch, writing custom **Batch Normalization** layers and an **Adam optimizer** from scratch.
- Benchmarked LeNet-5, ResNet-18, and ConvNeXt-tiny on the MNIST dataset, using bootstrap resampling to establish 95% confidence intervals for accuracy and error rates.

**Key Technologies:** PyTorch, NumPy, Numba JIT, Custom Backpropagation, Model Benchmarking, Bootstrap Statistics.

01/01/2025 – 31/01/2025

**Predictive Health Analytics & Stress Modeling**

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**Context:** Data Wrangling Project | Vrije Universiteit Amsterdam  
**Grade:** 9.0 / 10 (6.0 ECTS)

- Cleaned, transformed, and merged multi-source FitBit smartwatch and lifestyle datasets from Kaggle, resolving missing values and structural inconsistencies.
- Executed feature selection and correlation analyses to evaluate the influence of sleep stage ratios (deep, light, REM) and caffeine intake on sleep efficiency.
- Trained and evaluated machine learning classifiers (Decision Trees, Linear Regression) to predict daily stress categories based on sleep data and heart rate variability.
- Written visualization scripts for a professional presentation of the results, and analysis of trained models.

**Key Technologies:** Python, Pandas, Scikit-Learn, Feature Engineering, Data Cleaning, Decision Trees, Data Visualization (seaborn, matplotlib).

● **EDUCATION AND TRAINING**

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04/09/2023 – 31/07/2026 Amsterdam, Netherlands

**BACHELOR OF SCIENCE IN ARTIFICIAL INTELLIGENCE | TRACK: INTELLIGENT SYSTEMS** Vrije Universiteit Amsterdam

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**GPA:** 7.8 / 10 (Where an 8 is "Very good" and a 9 is "Excellent")

**Selected Coursework & Marks:**

- Multi-Agent Systems (9.5/10) | Computational Thinking (9.5/10)
- Data Wrangling (9.0/10) | Automata & Complexity (7.5/10)
- Data Structures and Algorithms for AI (8.5/10) | Intelligent Systems (8.5/10)
- Applied Programming for AI (8.5/10) | Computational Intelligence (8.0/10)

**Level in EQF** EQF level 6

01/09/2025 – 30/01/2026 Delft, Netherlands

**MINOR IN ELECTRONICS FOR ROBOTICS** TU Delft Faculty Electrical Engineering

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**Selected Coursework & Marks:**

- Circuit Analysis (9.0/10) | Marsrover Project (8.5/10)
- Digital Systems (8.0/10) | Power Electronics for Robotics (8.0/10)

**Level in EQF** EQF level 6

30/01/2023 – 30/06/2023 Valencia, Spain

**ERASMUS EXCHANGE SEMESTER** Universitat Politècnica de València

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**Level in EQF** EQF level 6

Level in EQF EQF level 6

WORK EXPERIENCE

PROJECT MANAGER INTERN – PROSKE GMBH – 05/09/2022 – 03/02/2023 – MUNICH, GERMANY

- **Project Coordination:** Managed the pre-event planning and on-site execution for the **ARD EEMEA Abbott Sales Conference** in Istanbul.
- **Budget & Supplier Management:** Coordinated with external suppliers, negotiated services, and assisted in tracking project expenses to ensure budget adherence.
- **Data & Evaluation:** Conducted in-process and post-event evaluations, compiling reports on attendees, content and operational metrics for senior management and account clients.
- **Cross-Functional Collaboration:** Cooperated with the production, content creation, and logistics teams to ensure seamless project delivery, based on clients' needs.

F&B SPECIALIST – HOTEL JAKARTA AMSTERDAM (MALABAR) – 01/11/2023 – 31/08/2024 – AMSTERDAM, NETHERLANDS

**Operational Management:** Assisting the Bar Manager in daily operations, including stock management and floor management for a high-quality venue.  
**Inventory Control:** Responsible for stock-taking and inventory management, using data to forecast supply needs and minimize waste (efficiency optimization).  
**Process Improvement:** Streamlining bar preparation processes (batching ingredients) to increase service speed and consistency.

LANGUAGE SKILLS

Mother tongue(s): **POLISH**

Other language(s):

|         | UNDERSTANDING |         | SPEAKING          |                    | WRITING |
|---------|---------------|---------|-------------------|--------------------|---------|
|         | Listening     | Reading | Spoken production | Spoken interaction |         |
| ENGLISH | C1            | C1      | C1                | C1                 | C1      |
| SPANISH | B1            | B1      | B1                | B1                 | B1      |
| FRENCH  | B1            | B1      | B1                | B1                 | B1      |

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user